



Iterative Compositional Zero-Shot Generation

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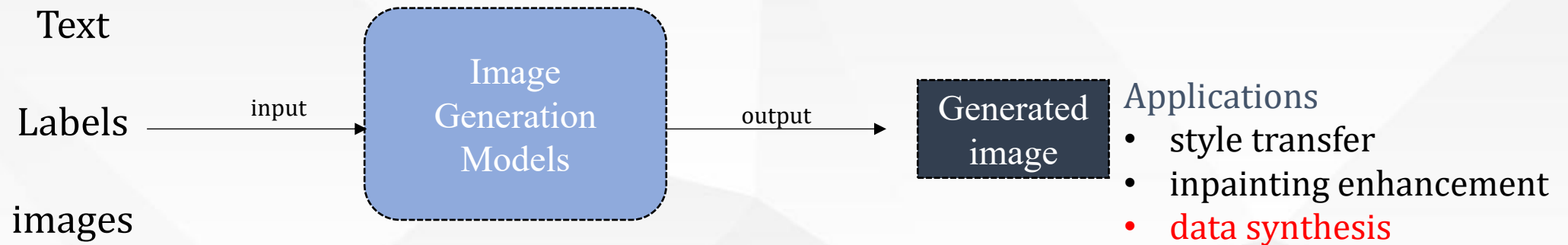
05 Conclusion

Part1

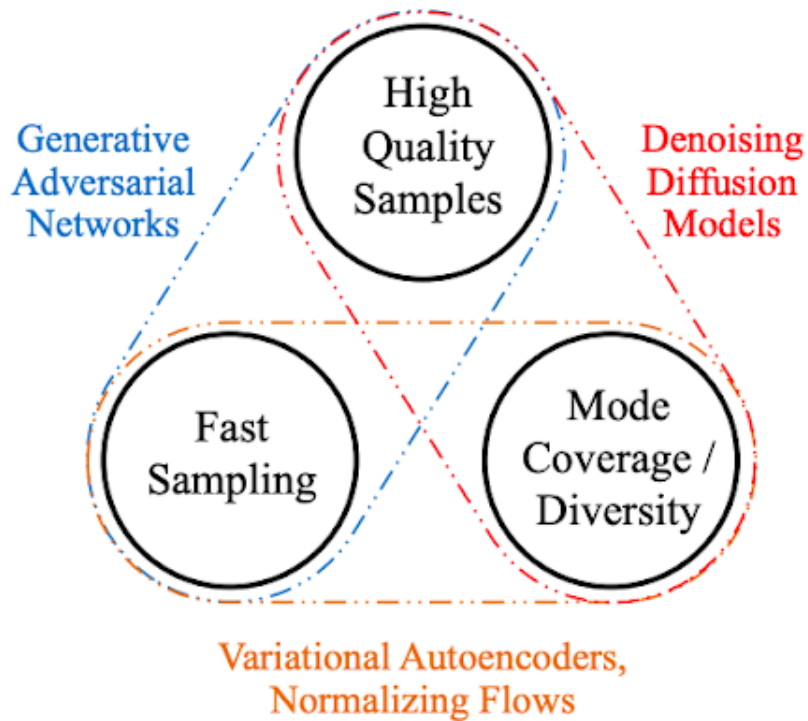
Introduction

Background

- In computer vision, image generation models—such as VAEs 、 GANs and diffusion models—are widely used. These models can generate diverse content from inputs like text 、 labels or images, with applications in style transfer 、 inpainting enhancement and data synthesis.



Background



VAE: Although training is stable, the generated images tend to be blurry and less detailed.

GAN: Produces sharper and more realistic images but often suffers from training instability and mode collapse.

Diffusion Models: Generate high-quality and diverse images by gradually denoising, becoming the current mainstream approach in image synthesis.

Limitations of Traditional UNet Architecture: Effective at capturing local image features but struggles with modeling long-range dependencies and complex global context.

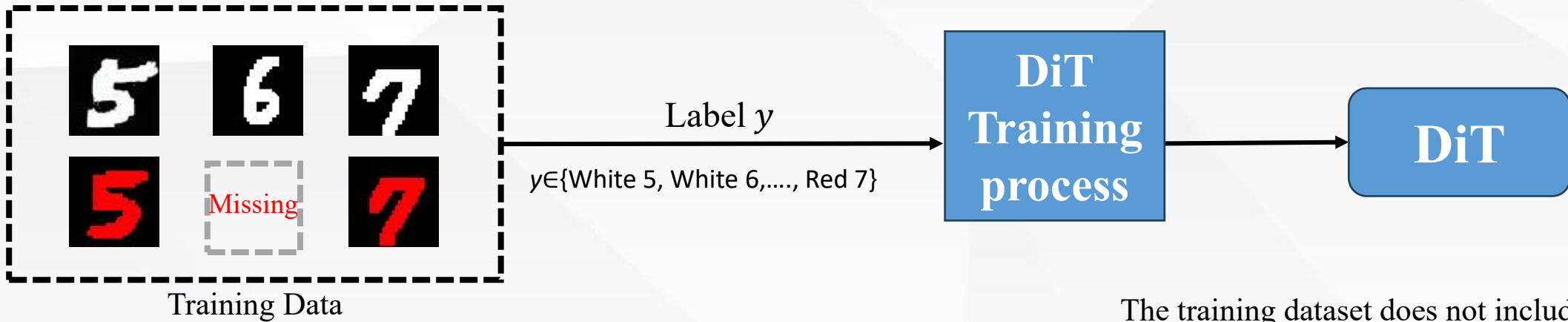
Advantages of Transformer Architecture: Excels at capturing global relationships, enhancing image diversity and resolution.

Therefore, our research use **Diffusion Transformer** (DiT)

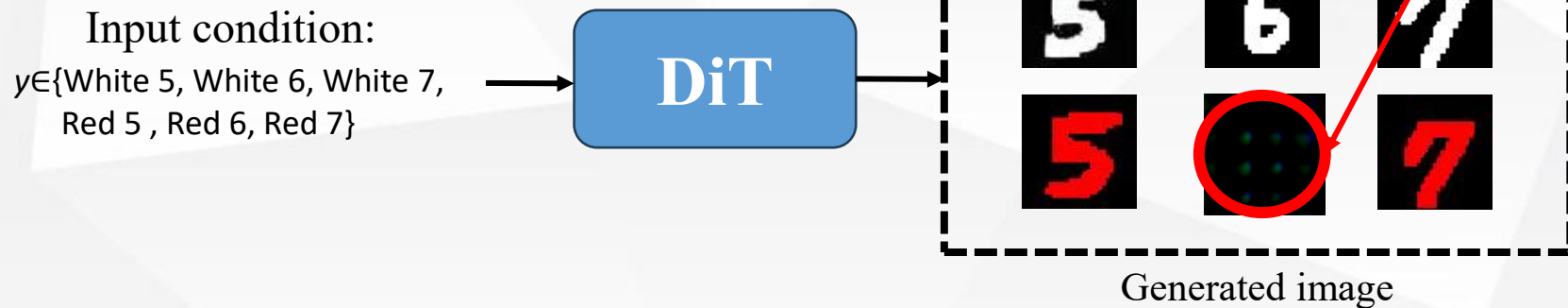
Research motivation

Limitation of Single-label Conditioning

- **Training:** train DiT with single label y



- **Inference Result**

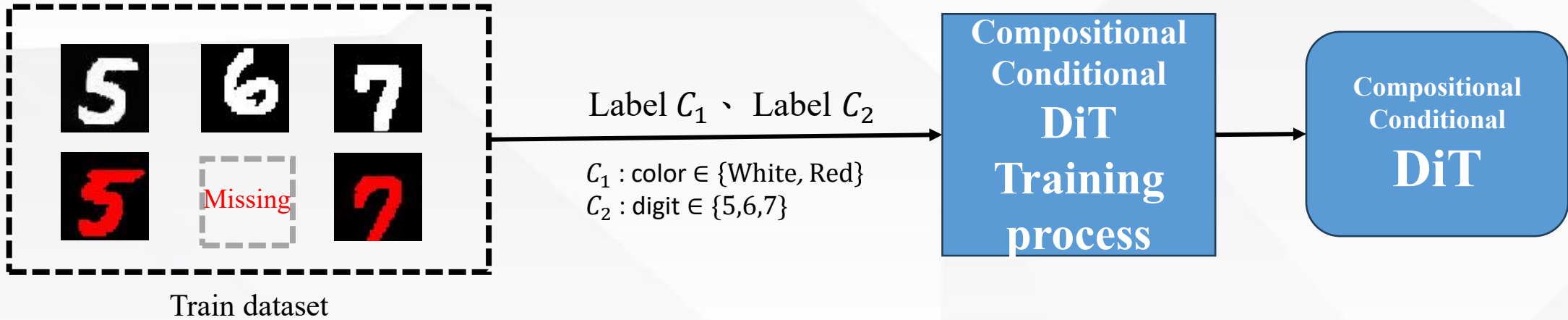


The training dataset does not include such samples, they can be generated, but their correctness is not guaranteed.

Research motivation

Our previous solution for the limitation

➤ Training: Compositional Conditional DiT

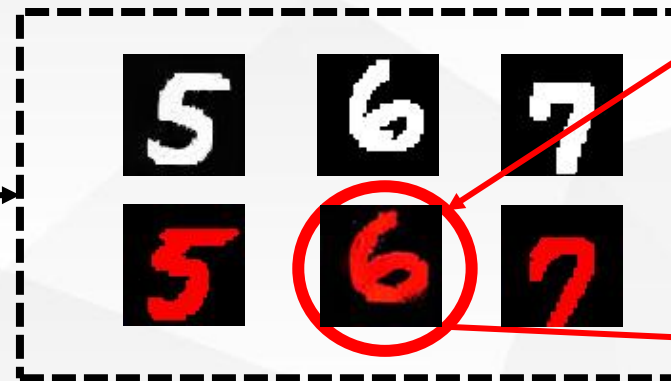
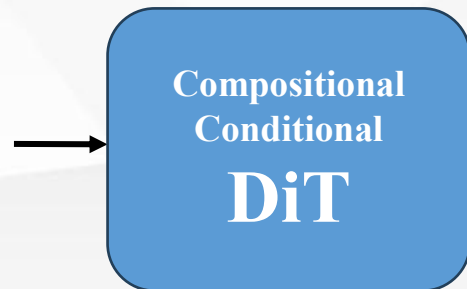


➤ Inference Result

Input condition:

$C_1 \in \{\text{White, Red}\}$

$C_2 \in \{5, 6, 7\}$



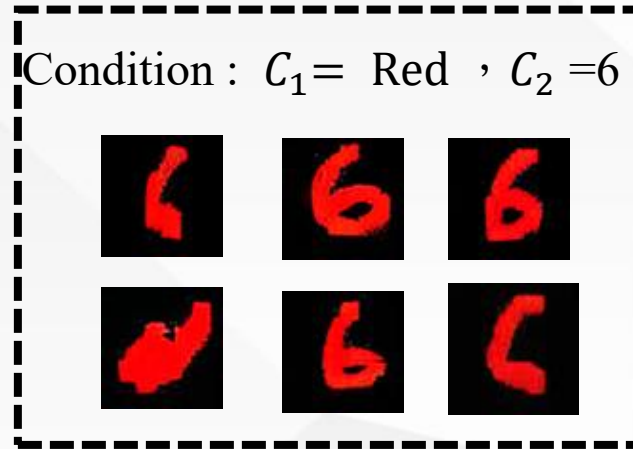
Generated image

Problem :
Some generated images are correct,
while others contain noticeable errors.

Research motivation

The problem of our previous solution

We randomly generate unseen images:

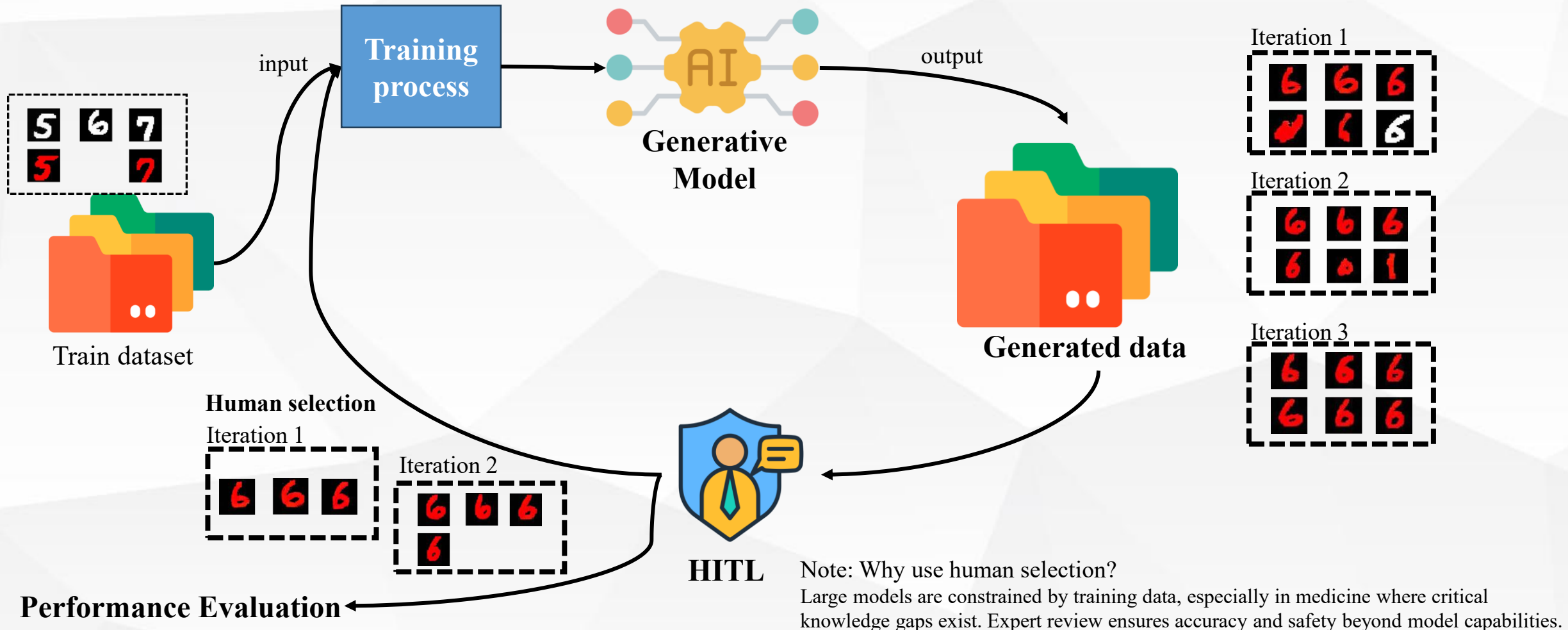


Compositional Conditional DiT may generate images that do not satisfy our purpose.
The ratio of good image may be low.

Research motivation

proposal → Learning from data + Human-in-the-Loop (HITL)

*Because generative AI models may produce hallucinated or factually incorrect content, human-in-the-loop validation is required to guarantee the accuracy of generated images.



Research motivation

Proposal challenge

Human-in-the-Loop verification requires manual inspection of each generated sample, which is labor-intensive and time-consuming.

- Limited manpower screening leads to **class imbalance** in the training data.

Possible Solutions

Traditional balanced resampling method for balanced training data

Our method

Modified balanced resampling achieves better generation performance in aforementioned iteration algorithm.

Contribution

1. Proposed a **compositional condition** training method within the Diffusion Transformer (DiT) architecture, enables flexible attribute composition and **generation of unseen images**.
2. Designed an iterative **data generation** framework with filtering, which progressively improves the accuracy of unseen image generation while maintaining high sample diversity.
3. Proposed a gamma-based modified balanced sampling strategy, where sampling weights are recalculated according to the updated class distribution in each iteration.

Part2

Related works

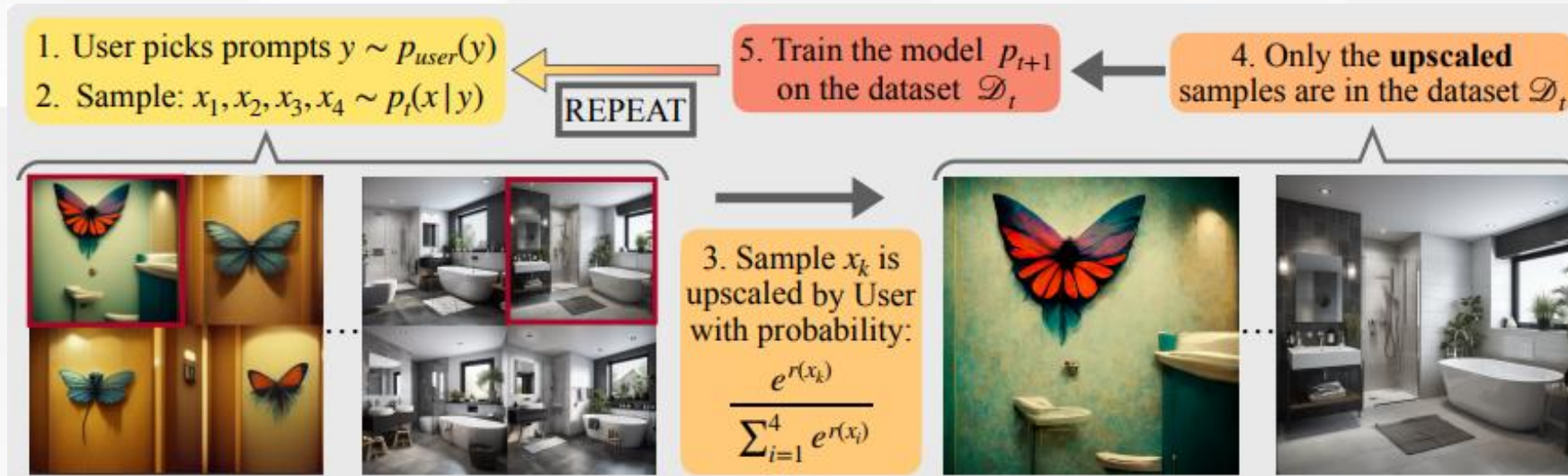
2-1 Iterative Retraining

2-2 imbalance data technology

2-3 Compositional Zero-Shot Learning & generation

2-1 Iterative Retraining

- Self-Consuming Generative Models with Curated Data (Ferbach et al., 2024)



The paper analyzes the impact of **data curation** on iterative retraining of generative models. It proves that when each training round uses samples curated by a reward model (such as human feedback or preference scores), the retraining process maximizes expected reward. At the same time, maintaining a portion of real data in each round ensures the stability of training.

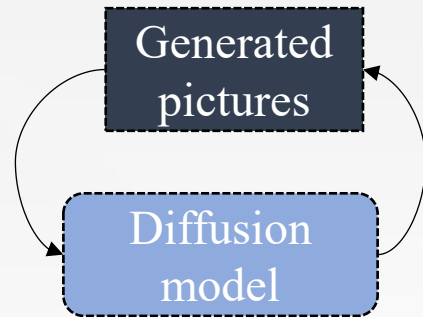
2-1 Iterative Retraining

- Self-Consuming Generative Models with Curated Data (Ferbach et al., 2024)

➤ Theorem 2.1 — Collapse Without Real Data

Pure Self-Consuming

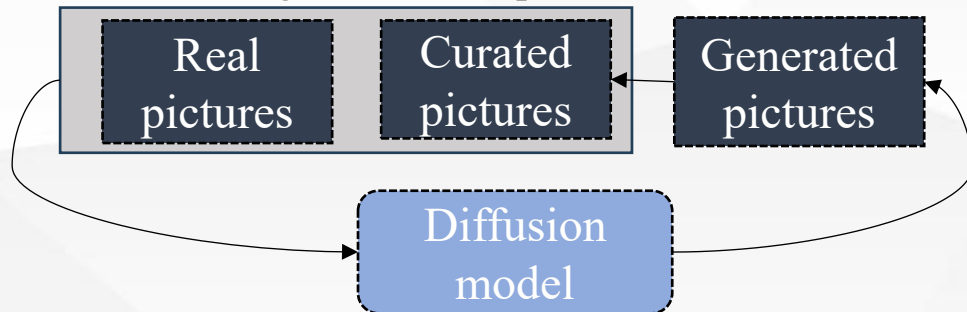
→ Mode Collapse to Highest Reward.



➤ Theorem 2.3 — Curation Increases Reward

Real + Curated Data

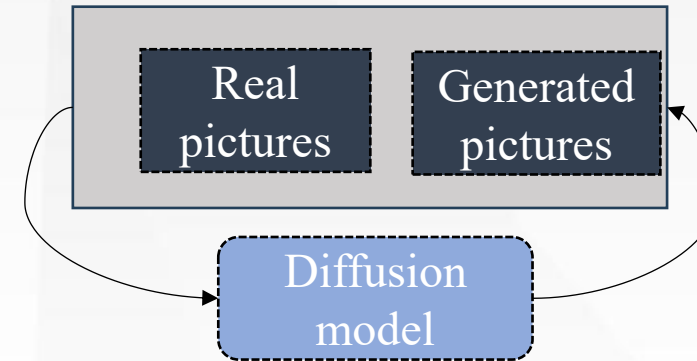
→ Stable Training + Reward Optimization



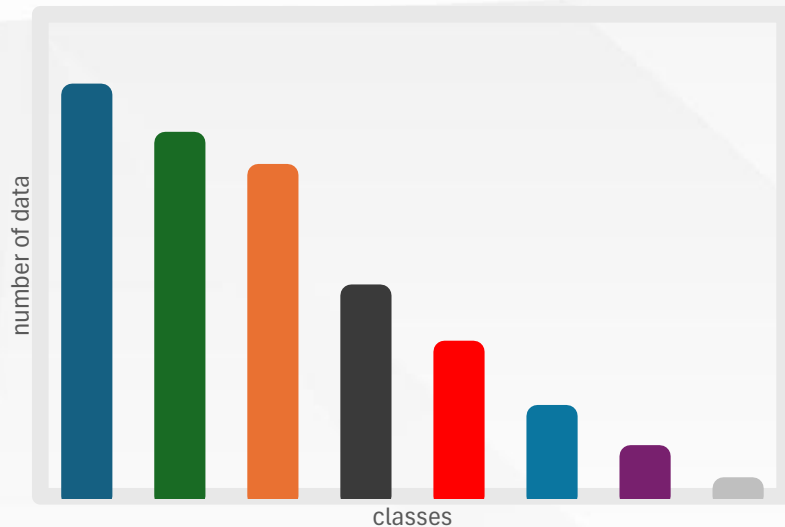
➤ Theorem 2.2 — Real Data Stabilizes Training

Real Data Mixture

→ Local Stability (prevents collapse)



2-2 Imbalance data technology



In real-world datasets, a few "Majority" classes dominate, while "Minority" or unseen compositional categories are extremely scarce.

- **Resampling**

Over-sampling: Repeatedly sampling minority data

→ Leads to Over-fitting on synthetic artifacts.

Under-sampling: Under-sampling: Removing majority data

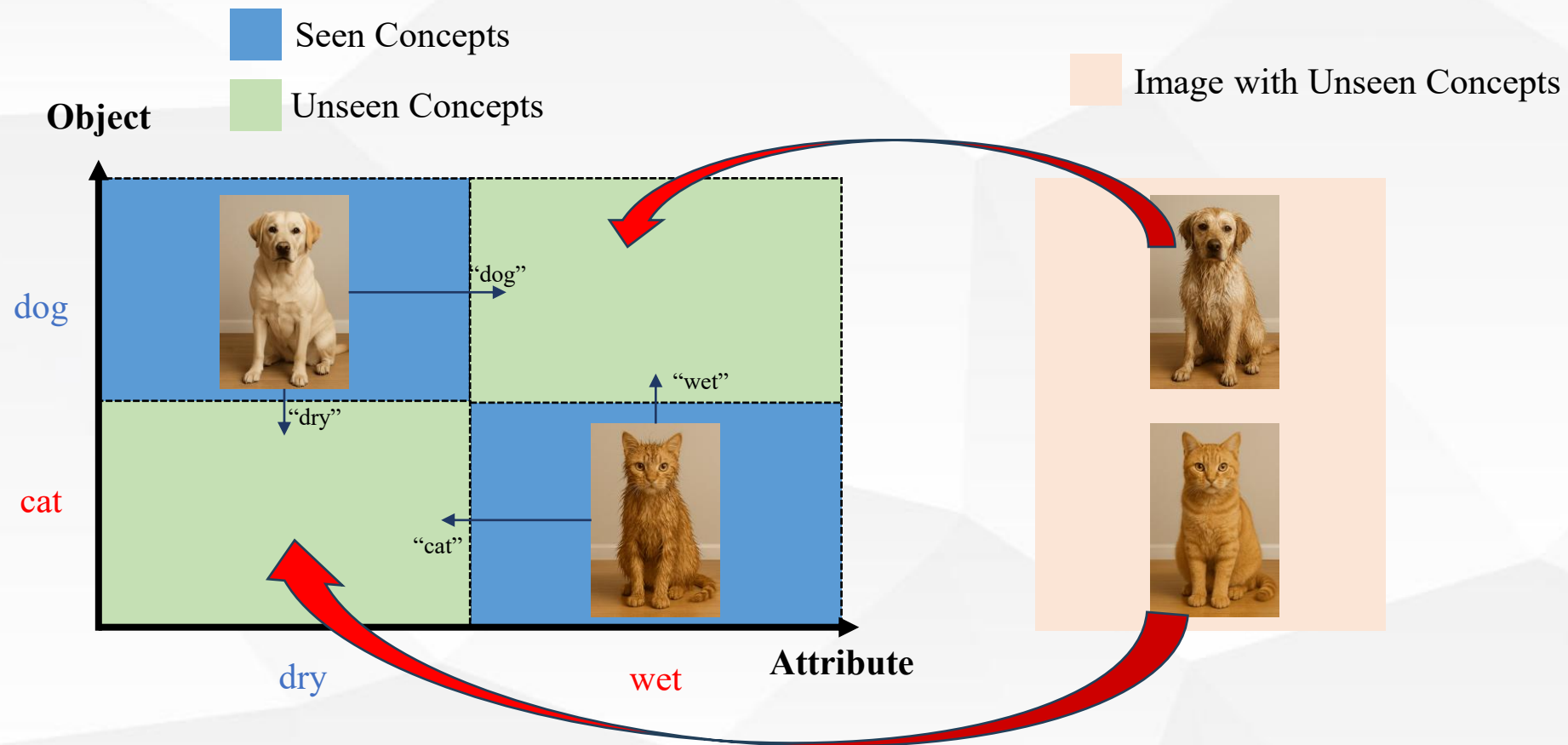
→ Results in Loss of General Features and diversity.

- **Reweighting**

Adjusting the loss function based on class frequency

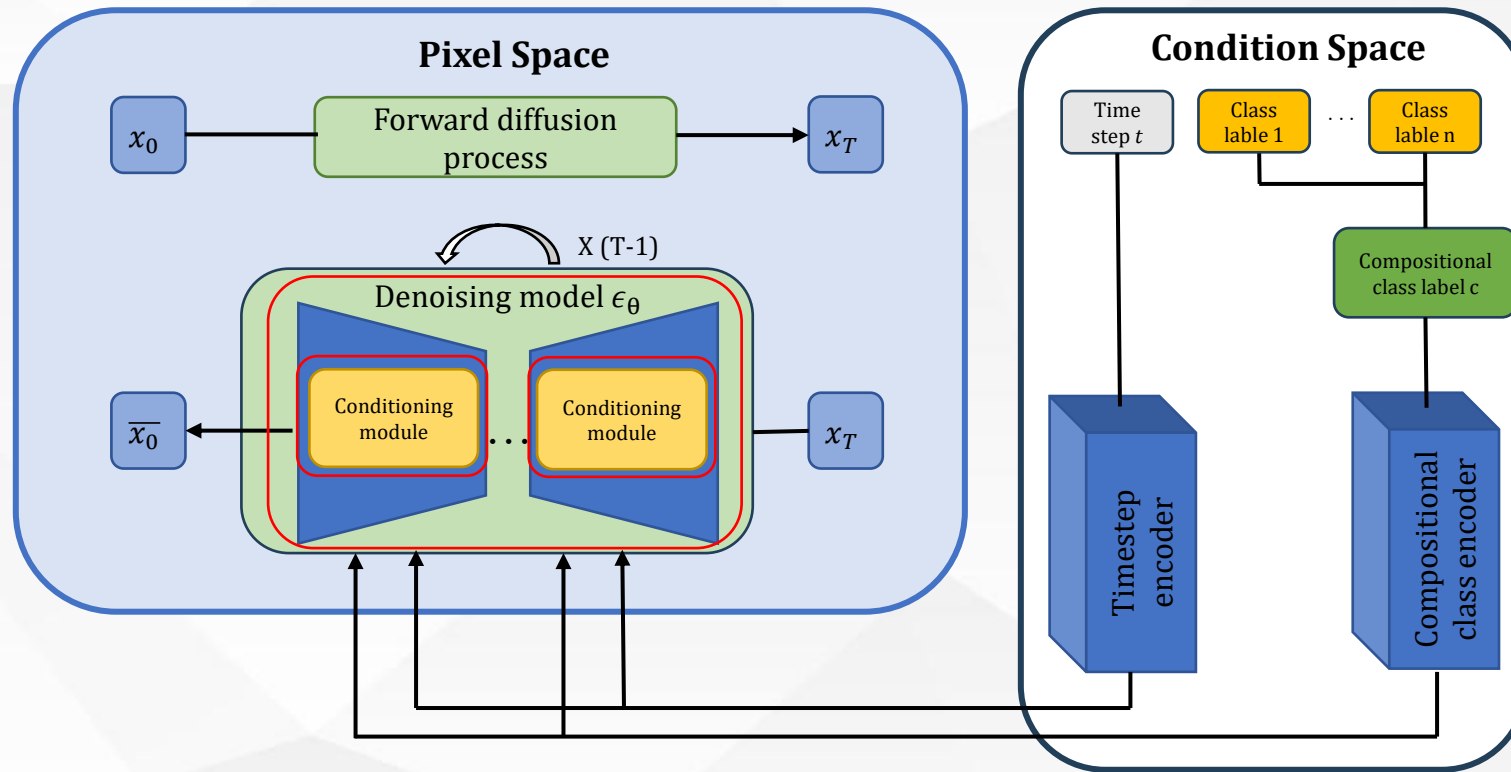
→ Difficult to stabilize in iterative generative loops.

2-3 Compositional Zero-Shot Learning & generation: Compositional Zero-Shot Learning



2-3 Compositional Zero-Shot Learning & generation: Compositional Conditional Diffusion Models

- Zero shot generation



- In our method ,
we replace the U-Net backbone with a **Transformer** backbone.
+ iterative retraining +modified balanced resampling

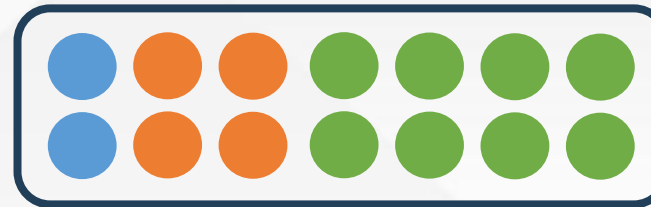
Part3

Method

- 3-1 Sampling Strategy**
- 3-2 Compositional DiT Model**
- 3-3 Data diversity evaluation**

3-1 Sampling Strategy (1/2)

: Why we need balance?



Initial dataset

Assume the dataset contains three categories (classes).

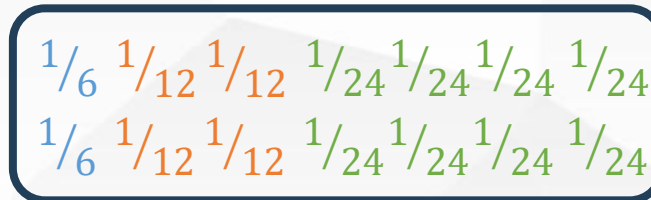
**Class-Probability
calculation**



Probability

$p = \text{probability} / \text{probability.sum()}$

$\sum p_i = \text{Number of Classes}$

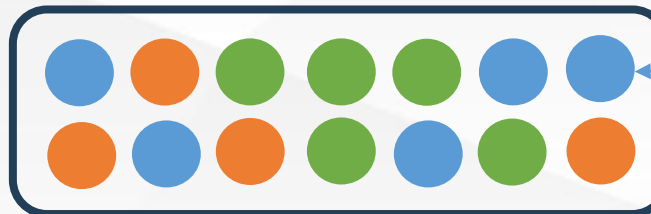


Random sampling with p



Note:

Repeated samples → Overfitting risk



Dataset after sampling

3-1 Sampling Strategy (2/2)

: Proposed Modified Balanced Sampling

- **Smoothed Probability** with γ

*N: number of samples per class

Gamma (γ) Value	Type	Probability Formula $p_i = 1/N_i^\gamma$	Outcome
$\gamma = 1$	Balanced sampling	$p_i = 1/N_i$	Strongest boost to rare classes .Most aggressive over-sampling .
$0 < \gamma < 1$	Smoothed Balanced sampling	$p_i = 1/N_i^\gamma$	A compromise that gently boosts rare classes. Helps prevent overfitting.
$\gamma = 0$	Random sampling	$p_i = 1/N_i^0 = 1$	All samples have equal probability. This is Simple uniform Sampling.

↓
Final Sampling Probability : $p = \text{probability} / \text{probability.sum()}$

↓ Random sampling with p

Dataset after sampling

3-1 Sampling Strategy (2/2)

: Proposed Modified Balanced Sampling

Algorithm1 : Modified Balanced Sampler

Input: Dataset $\mathcal{D}$ of size N , labels L corresponding to each sample in $\mathcal{D}$,

Smoothing parameter γ , seed s , epoch e

Output: An iterator of N sample indices.

Initialize:

- a. Count Class Frequency: Calculate `class_counts` for each unique class in L .
- b. Calculate Sample probabilities:
 - i. For each sample i with class label c :
$$\text{prob}_i \leftarrow 1 / (\text{class_counts}[c] ^ \gamma)$$
 - ii. Store all probabilities in a list P .
- c. Normalize Probabilities: Create a probability distribution $P \leftarrow P / \text{sum}(P)$.

Generate Indices:

- a. Set Seed: Use a seed for the current epoch ($\text{seed} = s + e$) to ensure reproducibility
- b. Weighted Random Sampling: Generate a list indices by sampling N indices from $\{0, \dots, N-1\}$ with replacement, using the probability distribution P .
- c. Shuffle (Optional): If enabled, shuffle the indices list to further randomize the order.
- d. Return an iterator for the indices list.

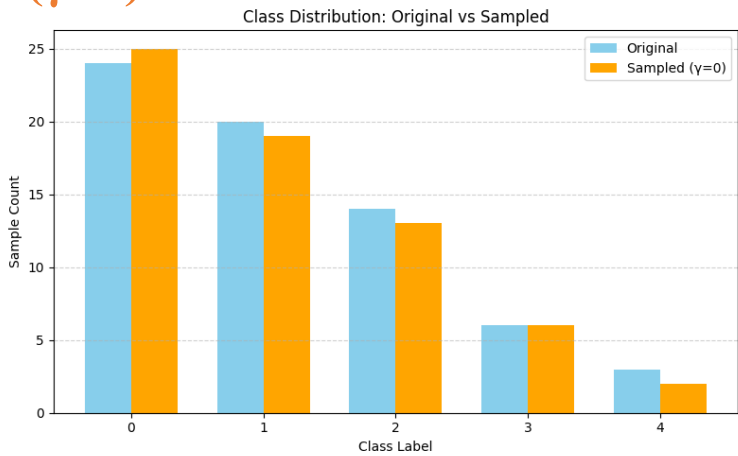
In our experiments, we evaluate different balancing strengths by setting $\gamma = 0, 0.3, 0.5, 0.7$ and 1 .

3-1 Sampling Strategy (2/2)

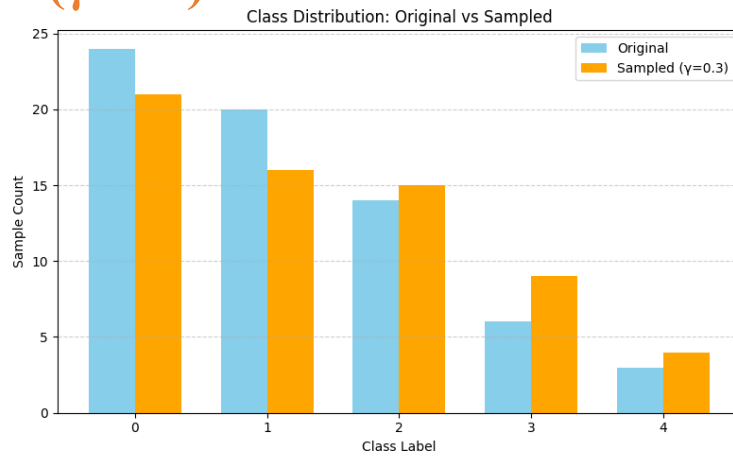
: Proposed Modified Balanced Sampling

- Visualizing Class Sampling Distribution under Different Gamma (γ) Values

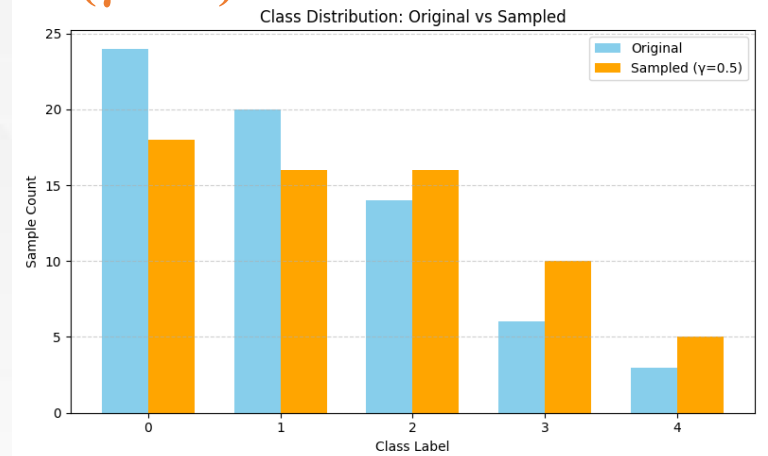
($\gamma=0$)



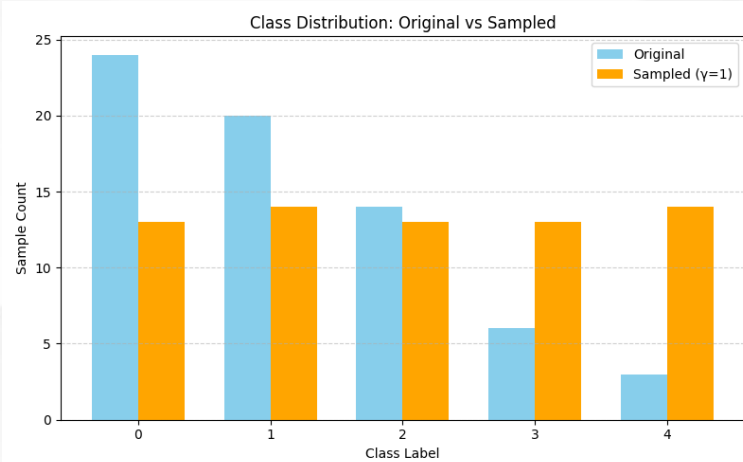
($\gamma=0.3$)



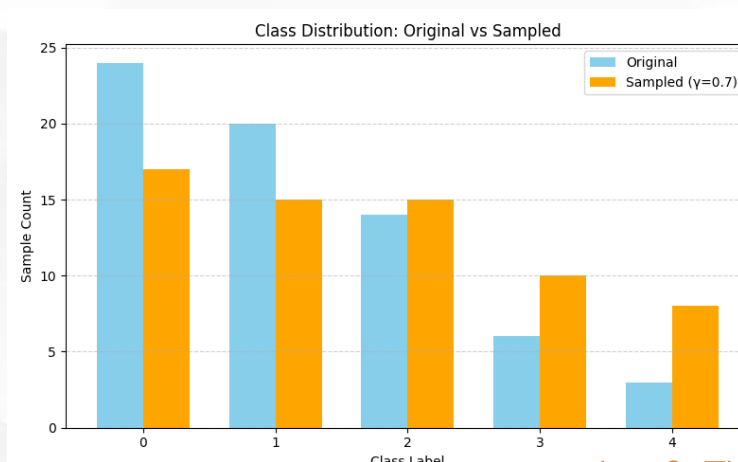
($\gamma=0.5$)



($\gamma=1$)

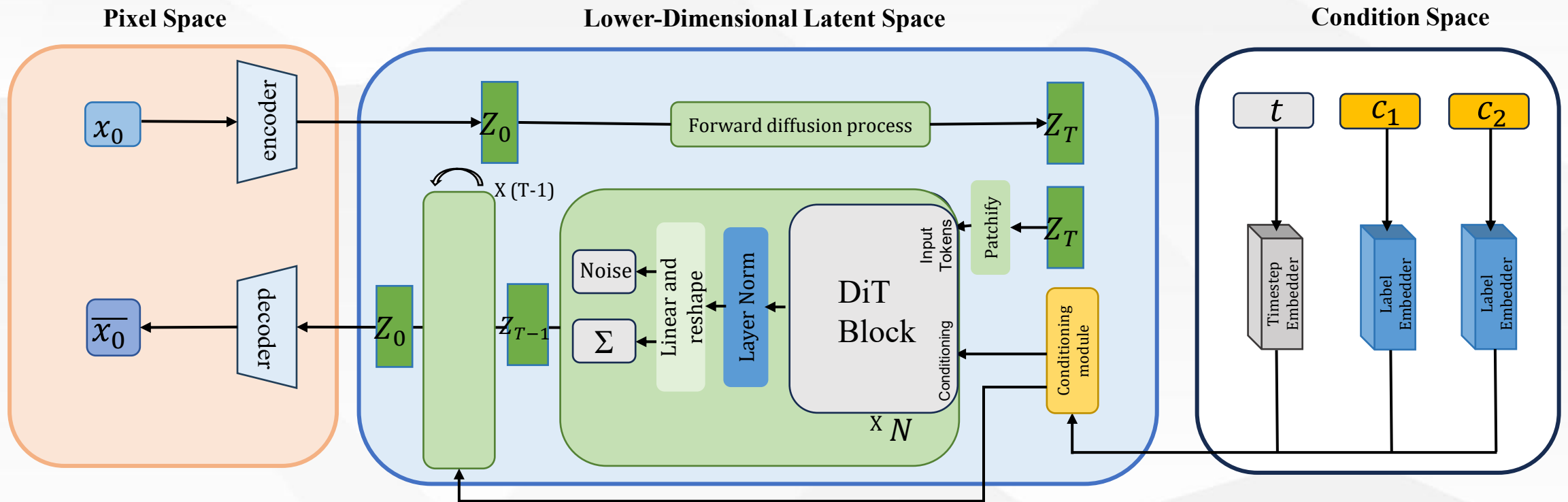


($\gamma=0.7$)



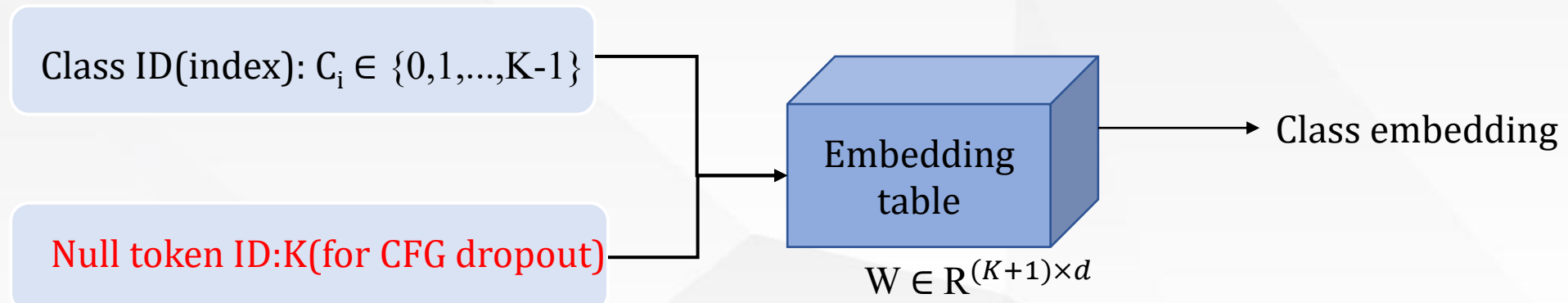
3-2 Compositional DiT Model

: Model Architecture Overview



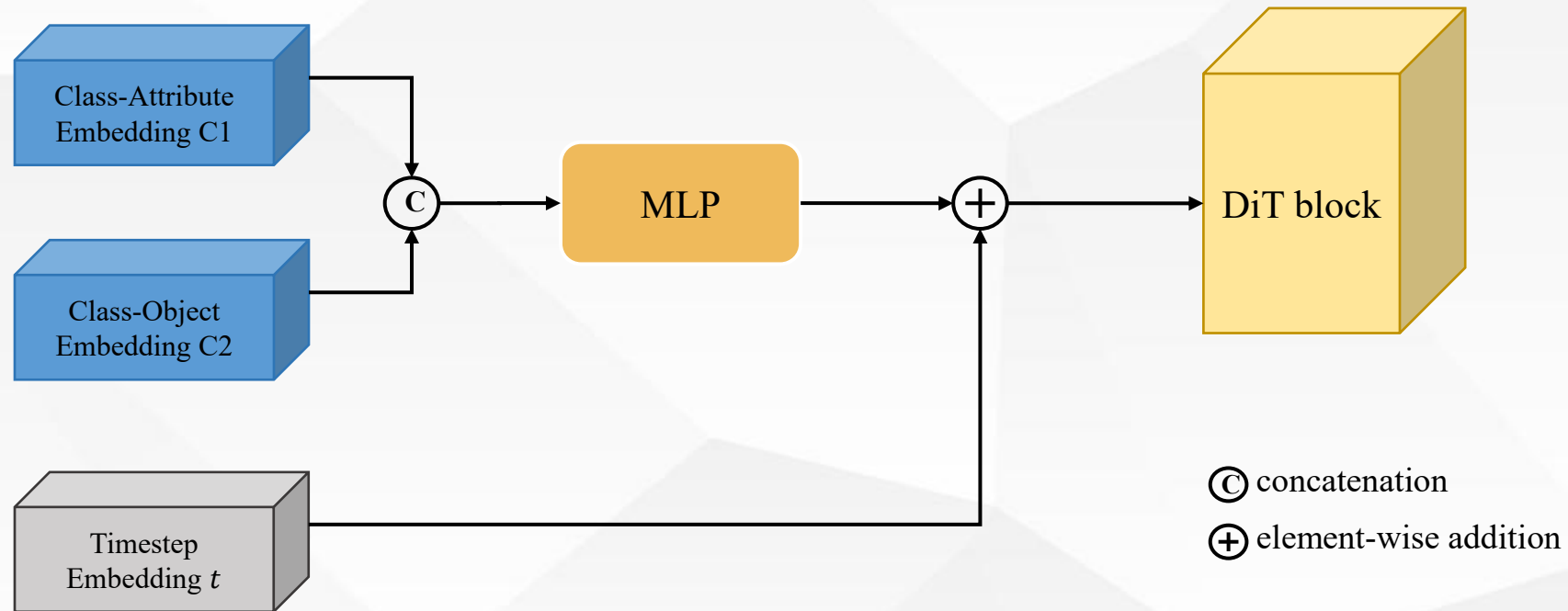
3-2 Compositional DiT Model

: Label Embedder



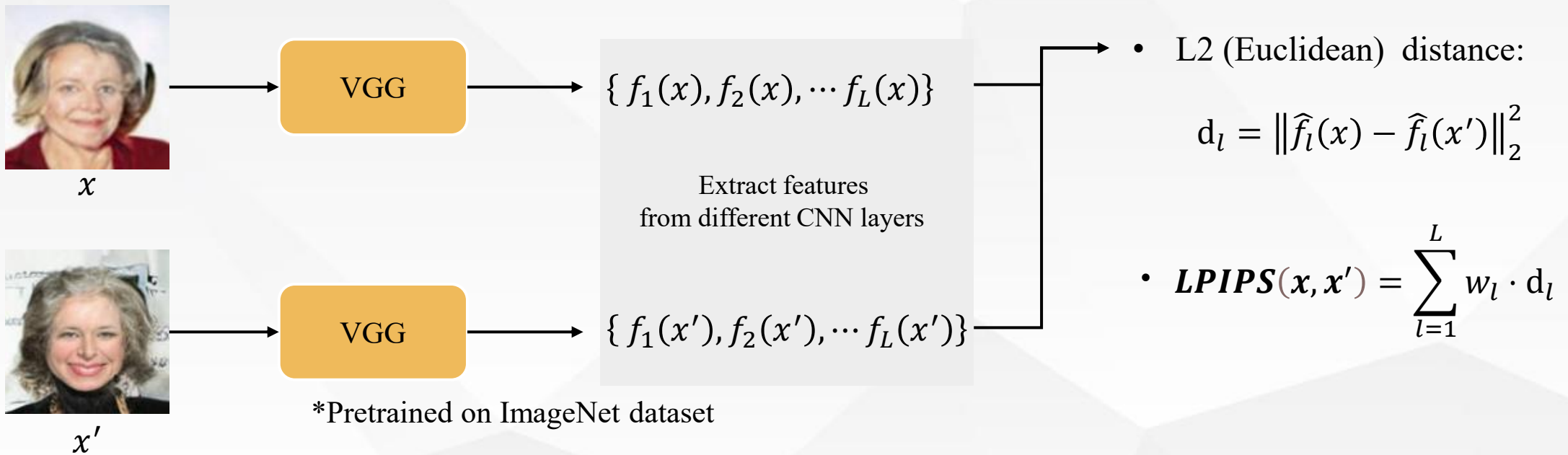
3-2 Compositional DiT Model

: Conditioning module



3-3 Data diversity evaluation

Evaluation metrics for **diversity**-Learned Perceptual Image Patch Similarity(LPIPS)



Part4

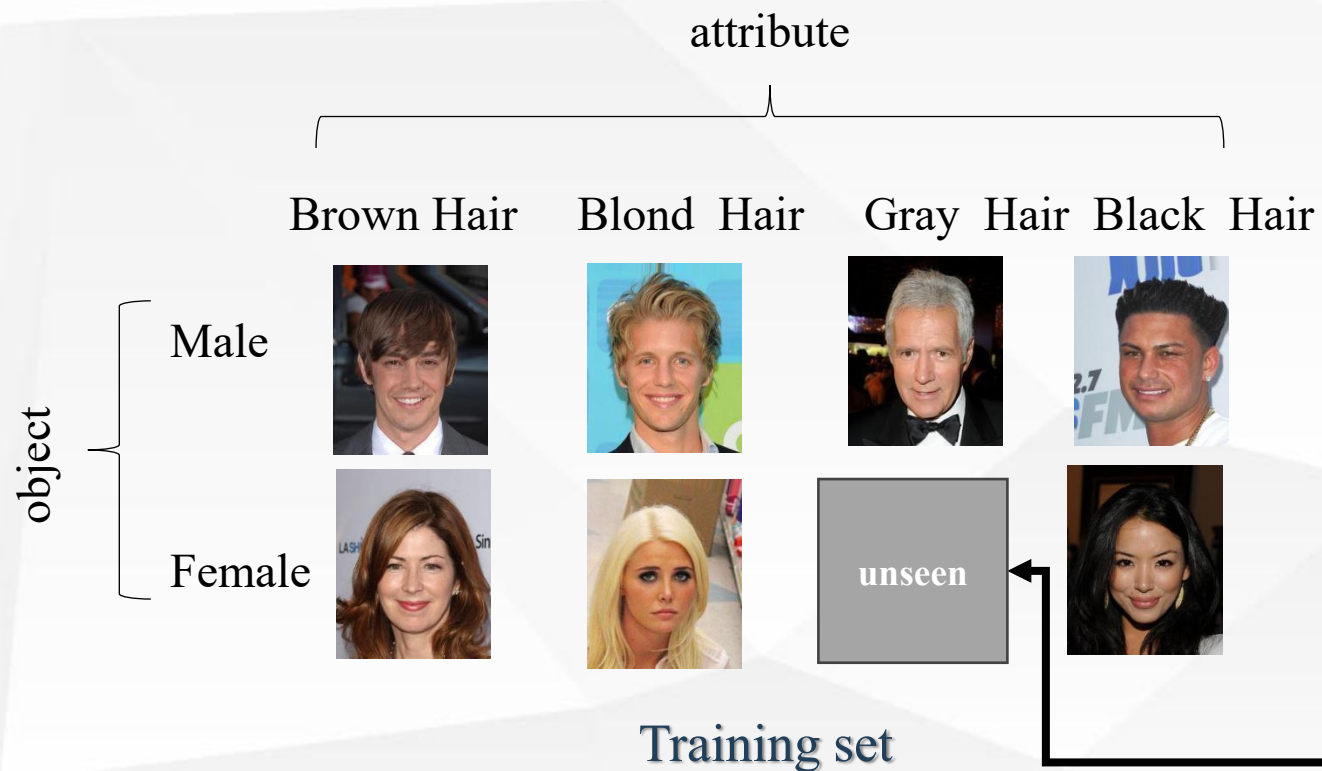
Experiment result

- 4-1 Experiment Setup**
- 4-2 Evaluation metrics**
- 4-3 Experiment results**

4-1 Experiment Setup:

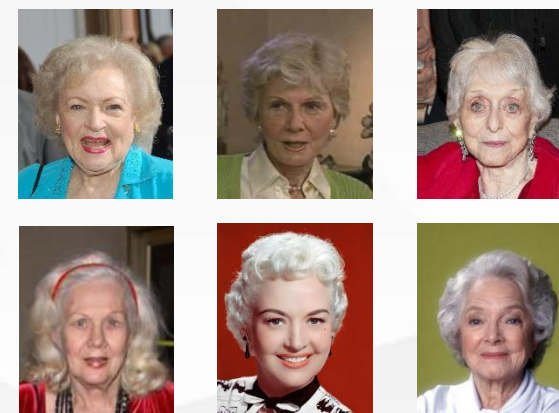
CelebA (CelebFace Attributes Dataset)

- CelebA is a large-scale public face dataset, containing over 200,000 celebrity images. Each image is annotated with 40 binary attribute labels (e.g., gender, hair color, smiling).



Target for Generation

<Gray Hair, Female>



Gray-Hair Female is excluded from the training set and treated as an **unseen** combination.

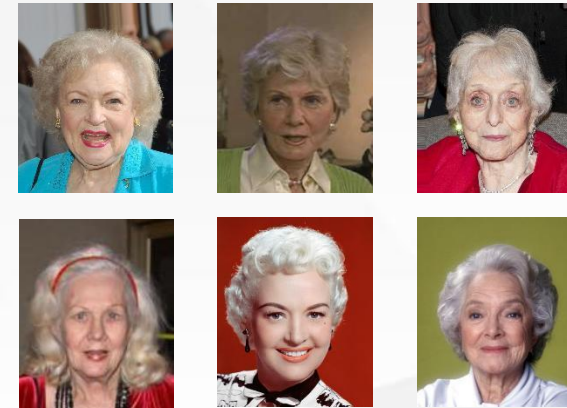
4-1 Experiment Setup: CelebA (CelebFace Attributes Dataset)

		attribute			
		Brown Hair	Blond Hair	Gray Hair	Black Hair
object	Male	12638	1676	6644	19621
	Female	20900	24358	0	19960

Training set

Target for Generation

<Gray Hair , Female>



Gray Hair Female is excluded from the training set and treated as **unseen**.

4-1 Experiment Setup:

CelebA (CelebFace Attributes Dataset)

- We generate images from unseen target attribute combinations and manually verify whether they match the target condition.

Pass



Fail (Does not match the target condition)



Selected images are added to the training set for the next

*For $\gamma = 0.3, 0.5$, and 0.7 , we omit the Iteration 1 experiment because unseen images have not yet been added, which makes class imbalance less severe at this stage.

4-2 Evaluation metrics

How do we evaluate ?

➤ Quantitative Analysis

The correctness ratio = $\frac{\text{number of manually selected correct images}}{\text{number of randomly generated unseen-type images}}$

Evaluation metrics for diversity : $LPIPS(x, x') = \sum_{l=1}^L w_l \cdot d_l$

In addition to the quantitative evaluation above, we also use t-SNE for qualitative analysis.

➤ Qualitative Diversity Evaluation

Step 1. Resize all generated images to 128×128

Step 2. Extract visual features using the intermediate layer of an ImageNet-pretrained VGG16

Step 3. Obtain a feature vector for each generated image

Step 4. Project high-dimensional features into 2D using t-SNE

Step 5. Visualize different iterations or γ values with different colors valuation above

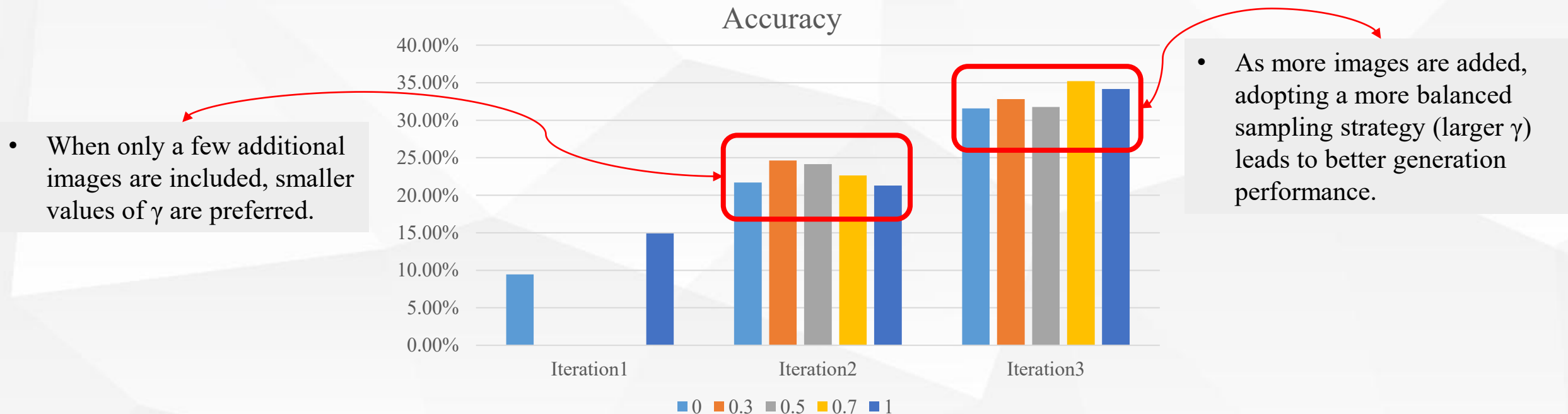
4-3 Experiment results

Quantitative Analysis of Iterative Training on CelebA Dataset

*Note:

- For $\gamma = 0.3$ 、 0.5 、 0.7 , we start from **Iteration 2**, using Iteration 1 ($\gamma = 1$) output as input
- The training dataset for **Iteration 3** is constructed from the updated dataset after image generation by each γ in **Iteration 2**.

	Random Sampling($\gamma=0$)	Modified Balanced Sampling($\gamma=0.3$)	Modified Balanced Sampling($\gamma=0.5$)	Modified Balanced Sampling($\gamma=0.7$)	Balanced Sampling($\gamma=1$)
Iteration 1	473 (9.46%)	-	-	-	622 (12.44%)
Iteration 2	1085 (21.7%)	1231 (24.62%)	1123 (24.16%)	1132 (22.64%)	1065 (21.3%)
Iteration 3	1579 (31.58%)	1641 (32.82%)	1588 (31.76%)	1751 (35.2%)	1708 (34.16%)



4-3 Experiment results

• Quantitative Analysis of Iterative Training on CelebA Dataset

*Note:

- For $\gamma = 0.3$ 、 0.5 、 0.7 , we **start from Iteration 2**, using Iteration 1 ($\gamma = 1$) output as input
- The training dataset for **Iteration 3** is constructed from the updated dataset after image generation by each γ in **Iteration 2**.

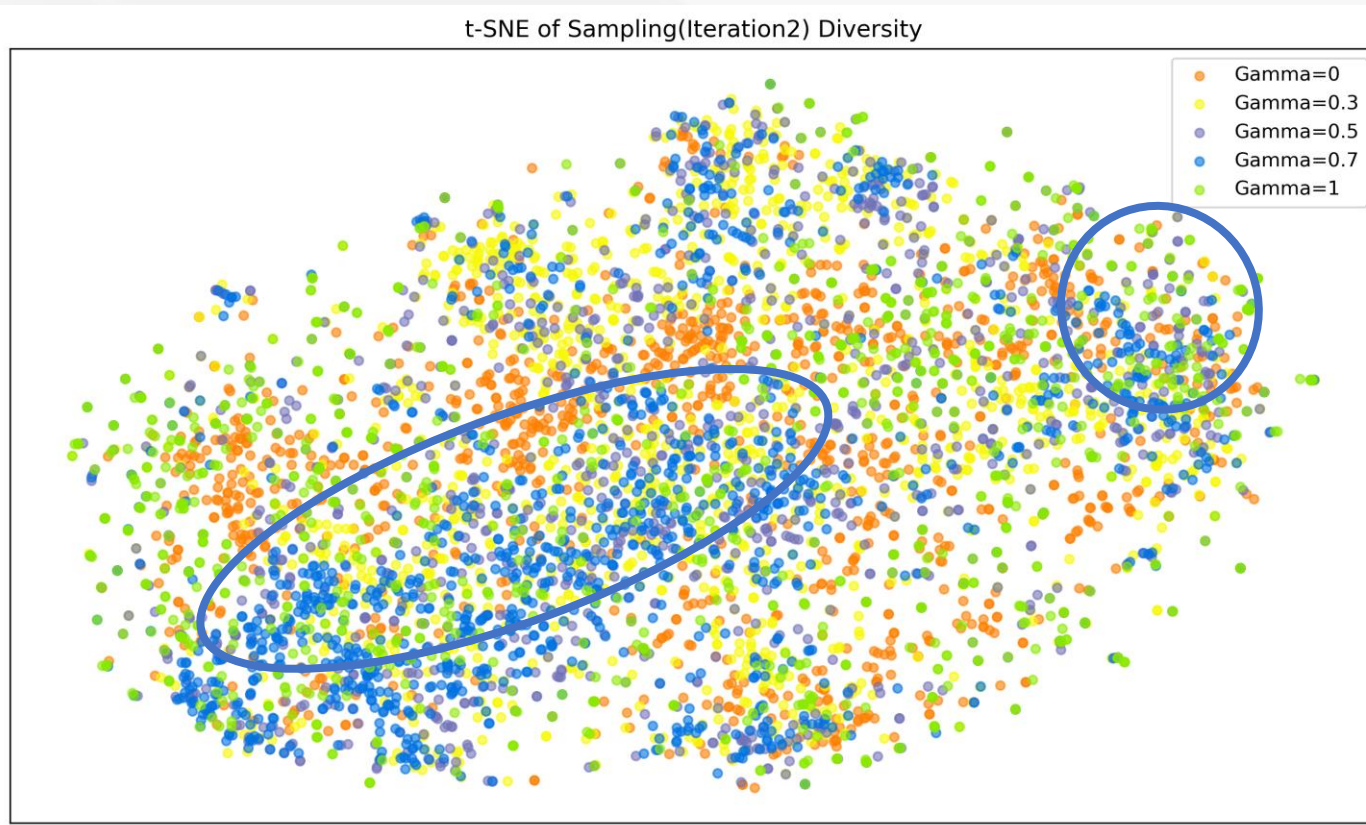
➤ LPIPS

	Random Sampling($\gamma=0$)	Modified Balanced Sampling($\gamma=0.3$)	Modified Balanced Sampling($\gamma=0.5$)	Modified Balanced Sampling($\gamma=0.7$)	Balanced Sampling($\gamma=1$)
Iteration1	0.5339	-	-	-	0.5596
Iteration2	0.5276	0.5346	0.5269	0.5082	0.5333
Iteration3	0.4461	0.4967	0.4675	0.4614	0.5117

4-3 Experiment results

• Qualitative Analysis of Iterative Training on CelebA Dataset

	Random Sampling($\gamma=0$)	Modified Balanced Sampling($\gamma=0.3$)	Modified Balanced Sampling($\gamma=0.5$)	Modified Balanced Sampling($\gamma=0.7$)	Balanced Sampling($\gamma=1$)
Iteration2	0.5276	0.5346	0.5269	0.5082	0.5333



a. t-SNE of Iteration2

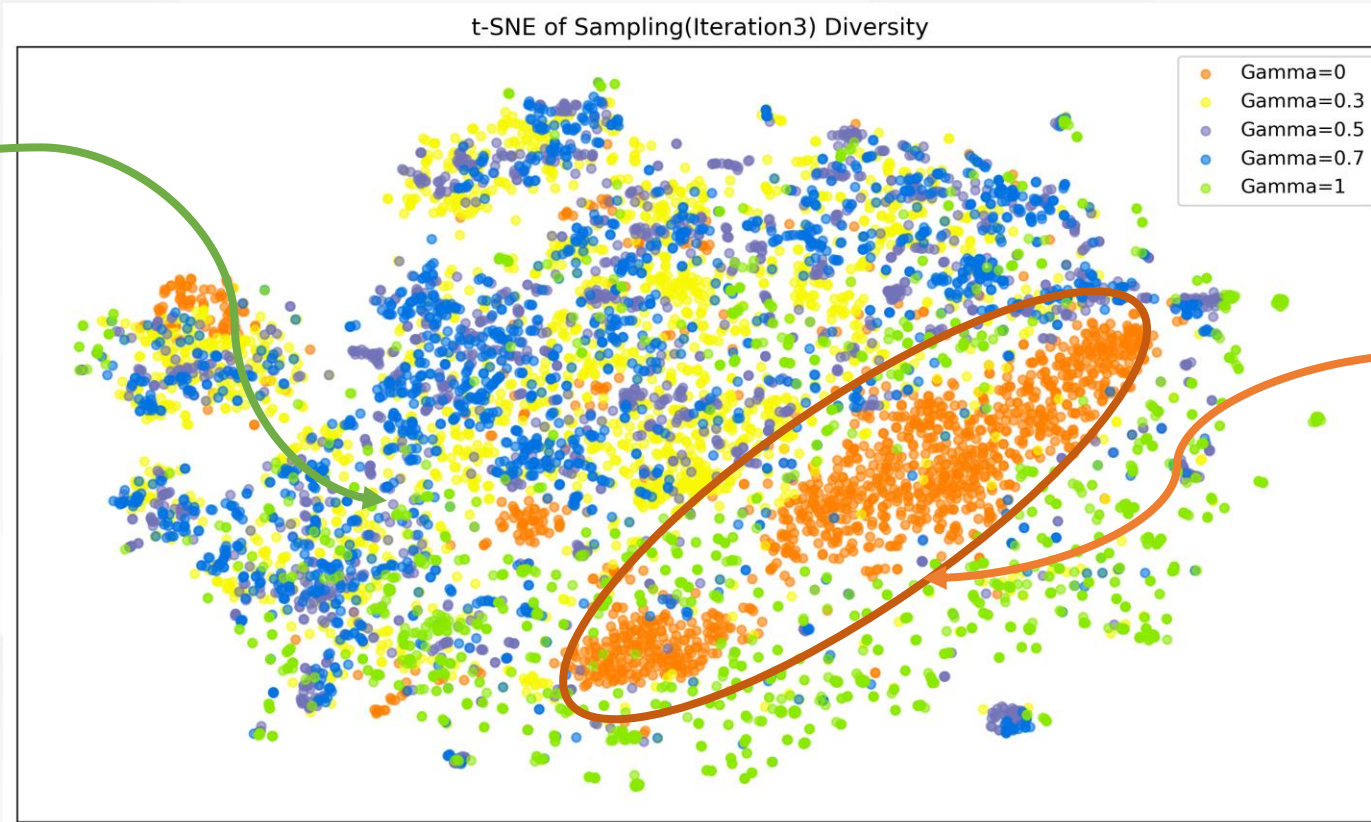
When the diversity score is higher, the t-SNE points are more spread out, whereas lower diversity scores lead to more compact clusters. In this figure, the case with $\gamma=0.7$ has a lower diversity score and appears noticeably more concentrated in the t-SNE plot.

4-3 Experiment results

• Qualitative Analysis of Iterative Training on CelebA Dataset

	Random Sampling($\gamma=0$)	Modified Balanced Sampling($\gamma=0.3$)	Modified Balanced Sampling($\gamma=0.5$)	Modified Balanced Sampling($\gamma=0.7$)	Balanced Sampling($\gamma=1$)
Iteration3	0.4461	0.4967	0.4675	0.4614	0.5117

$\gamma = 1$: Dispersed
(High Diversity)



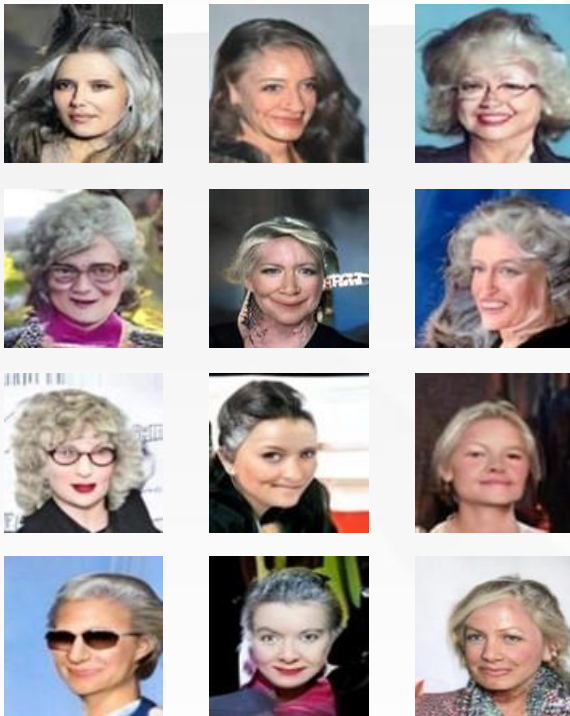
$\gamma = 0$: Concentrated
(Low Diversity)

b. t-SNE of Iteration3

4-3 Experiment results

Balanced Sampling($\gamma=1$)

<Gray Hair , Female>



1. Selected Images from **Iteration 1**

LPIPS:0.5596
Accuracy:12.44%

Balanced Sampling($\gamma=1$)

<Gray Hair , Female>



3. Selected Images from **Iteration 3**

LPIPS:0.5117
Accuracy:34.16%

By strategically reducing diversity, our method significantly boosts generation accuracy

Part5

Conclusion

This study presents an **iterative compositional DiT framework** for unseen attribute-object image generation. The model uses attribute and object conditions to guide latent diffusion with a Transformer backbone.

By combining **curated iterative retraining** and **gamma-based modified balanced sampling**, the proposed method progressively improves target-condition correctness while maintaining reasonable sample diversity. Experiments on CelebA show that gamma sampling provides a flexible way to balance generation accuracy and diversity under dynamic class imbalance.

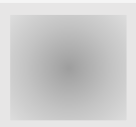
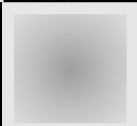
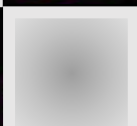
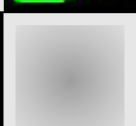
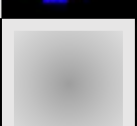


Thanks for listening!

Appendix

Experiment Setup: Mnist Color Dataset

- To demonstrate the model's ability to **generate unseen images**, we first conducted a preliminary test on a simpler toy dataset, where combinations along the diagonal (e.g., <Zero, White>) were held out as unseen data.

		attribute				
		0	1	2	3	4
object	White					
	Red					
	Green					
	Blue					
	Yellow					

Training set

Target for Generation

• <Zero, White>



• <One, Red>



• <Two, Green>



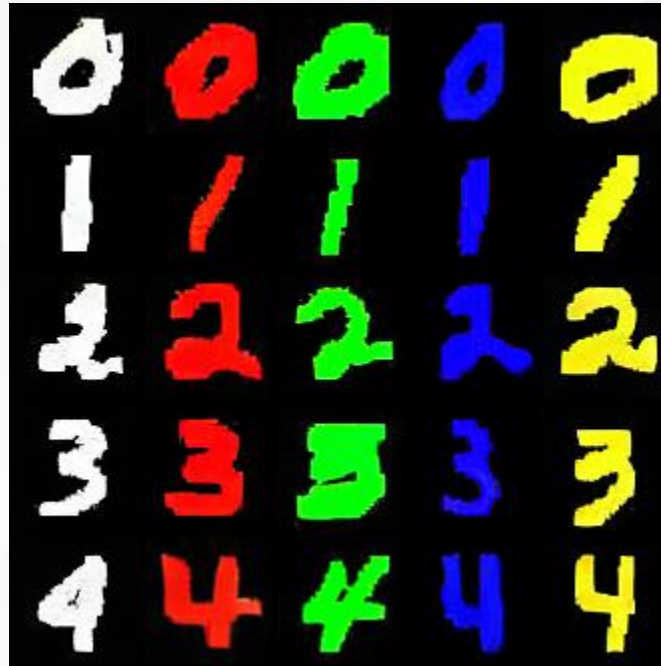
• <Three, Blue>



• <Four, Yellow>



Experiment Results: Mnist Color Dataset - sample

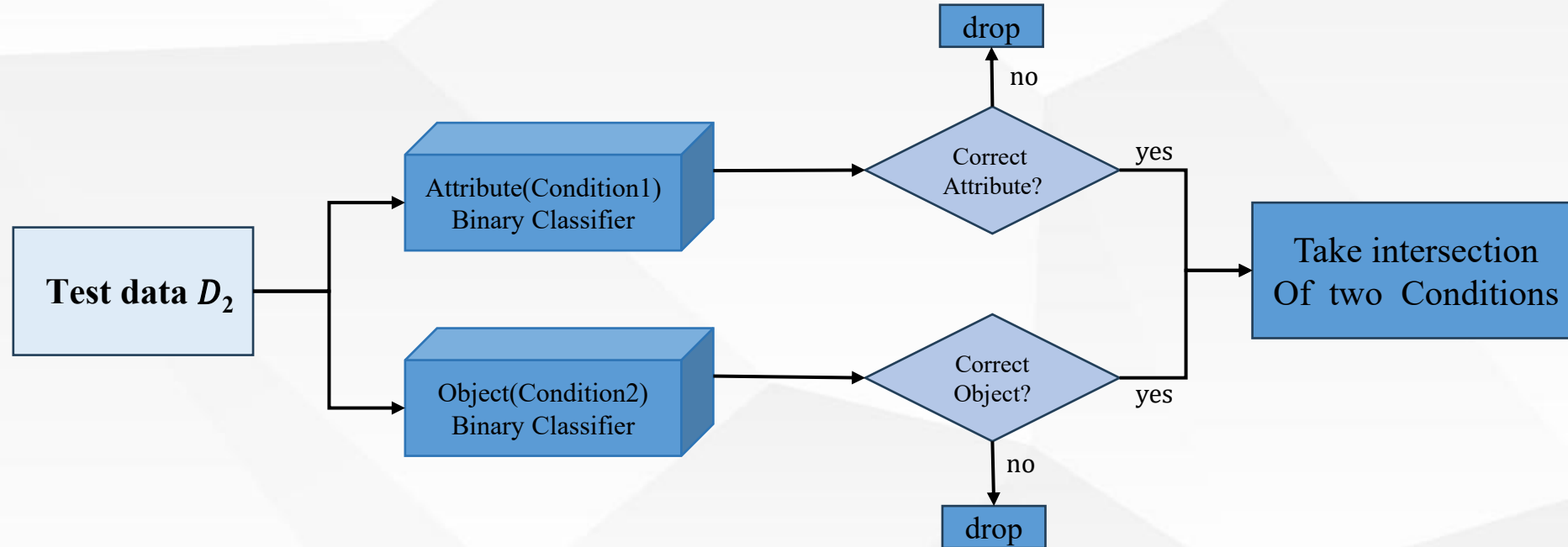


Classifier

- Ablation Study: Impact of Generated Data

Classifier 1: Real data only → Baseline performance

Classifier 2: Real + Generated → Improved accuracy by X%



Classifier

➤ Iteration 2

➤ Iteration 3

Use 1000 Gray_Hair Female images as Test data D_2

score classifier	Condition1 Number of corrects	Condition2 Number of corrects	Composition Number of corrects
Baseline	230	949	205
Balanced Sampling	286	935	244
Balanced Sampling	349	957	322
Uniform Sampling	280	951	245
Uniform Sampling	234	946	203
Modified Balanced Sampling (gamma=0.5)	381	939	339
Modified Balanced Sampling (gamma=0.5)	391	948	353
Modified Balanced Sampling (gamma=0.3)	361	929	310
Modified Balanced Sampling (gamma=0.3)	397	927	350
Modified Balanced Sampling (gamma=0.7)	354	932	313
Modified Balanced Sampling (gamma=0.7)	400	915	344